

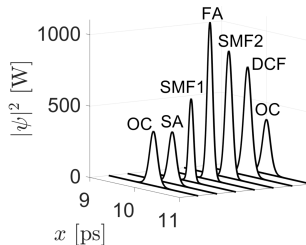
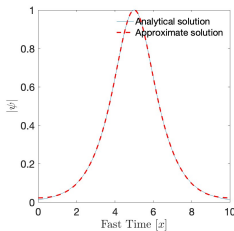


Stability in a short pulse laser

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Laser modeling

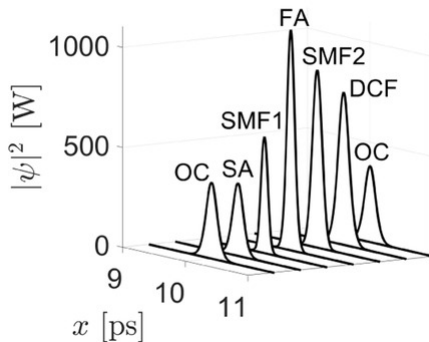
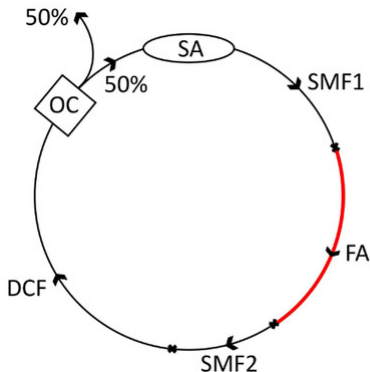


- 1 Fiber lasers generate **solitons**, or **breathers**¹ by balancing loss and gain, non-linearity and dispersion.
- 2 **Modeling goal:** Determine the region in system parameter space, where a stable pulse exists, and optimize the pulse parameters.

¹ Akhmediev, N. and A. Ankiewicz (2008) Dissipative solitons: from optics to biology and medicine. Springer Science Business Media.

Kim laser model

Our model of the lumped laser is a dispersion-managed laser by Kim. et al.²



²H. Kim, P. Qin, Y. Song, H. Yang, J. Shin, C. Kim, K. Jung, C. Wang, and J. Kim, Sub-20-attosecond timing jitter mode-locked fiber lasers, IEEE J. Sel. Top. Quantum Electron, 20 (2014), pp. 260-267. [□](#) [◀](#) [▶](#) [◀◀](#) [▶▶](#) [⌂](#) [↺](#) [↻](#)

Mathematical model

For our laser model, each component is modeled by a transfer function $\mathcal{P} : L^2(\mathbb{R}, \mathbb{C}) \rightarrow L^2(\mathbb{R}, \mathbb{C})$ of each component as

$$\psi_{\text{out}} = \mathcal{P}\psi_{\text{in}},$$

where ψ_{in} and ψ_{out} are the input and output pulses, respectively. The round trip operator $\mathcal{R}$ is modeled by

$$\mathcal{R} = \mathcal{P}^{\text{OC}} \circ \mathcal{P}^{\text{DCF}} \circ \mathcal{P}^{\text{SMF2}} \circ \mathcal{P}^{\text{FA}} \circ \mathcal{P}^{\text{SMF1}} \circ \mathcal{P}^{\text{SA}}.$$

The **monodromy operator** $\mathcal{M}$ is a linearization of the round trip operator $\mathcal{R}$, and is modeled by

$$\mathcal{M} = \mathcal{U}^{\text{OC}} \circ \mathcal{U}^{\text{DCF}} \circ \mathcal{U}^{\text{SMF2}} \circ \mathcal{U}^{\text{FA}} \circ \mathcal{U}^{\text{SMF1}} \circ \mathcal{U}^{\text{SA}},$$

where the linearized transfer function $\mathcal{U}$ is defined by

$$u_{\text{out}} = \mathcal{U}u_{\text{in}} = u(L_{\text{FA}}, \cdot)$$

Fast saturable absorber

The fast saturable absorber is modeled by a fast saturable loss transfer function $\mathcal{P}^{\text{SA}}$, given by

$$\psi_{\text{out}} = \mathcal{P}^{\text{SA}}(\psi_{\text{in}}) = \left(1 - \frac{l_0}{1 + |\psi_{\text{in}}|^2 / P_{\text{sat}}} \right) \psi_{\text{in}},$$

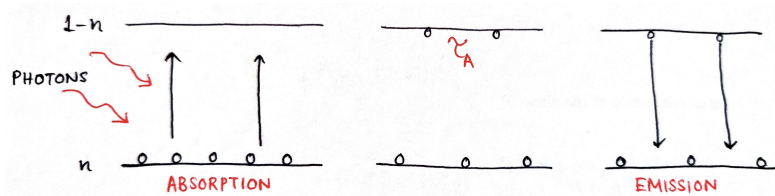
where l_0 is the unsaturated loss, and P_{sat} is the saturation power. Then, the linearized transfer function $\mathcal{U}^{\text{SA}}$ is given by

$$u_{\text{out}} = \mathcal{U}^{\text{SA}}(u_{\text{in}}) = \left(1 - l(\psi_{\text{in}}) - \frac{2l^2(\psi_{\text{in}})}{l_0 P_{\text{sat}}} \psi_{\text{in}} \psi_{\text{in}}^T \right) u_{\text{in}},$$

where

$$l(\psi_{\text{in}}) = \frac{l_0}{1 + |\psi_{\text{in}}|^2 / P_{\text{sat}}}.$$

Two-level energy model of SESAM



- $n = n(\psi)$ is the fraction of the population in the lower level (ground state).³
- $1 - n$ is the population in the upper level (excited state).

³Wang, Shaokang, et al. "Obtaining more energetic modelocked pulses from a SESAM-based fiber laser." Optics Express 28.14 (2020): 20345-20361.

Mathematical formulation

We model the SESAM using the transfer function $\mathcal{P}^{\text{SA}}$ in terms of saturable gain coefficient ρ as

$$\psi_{\text{out}} = \mathcal{P}^{\text{SA}}\psi_{\text{in}} = \exp(-\rho n(\psi_{\text{in}}))\psi_{\text{in}}, \quad \rho > 0,$$

where

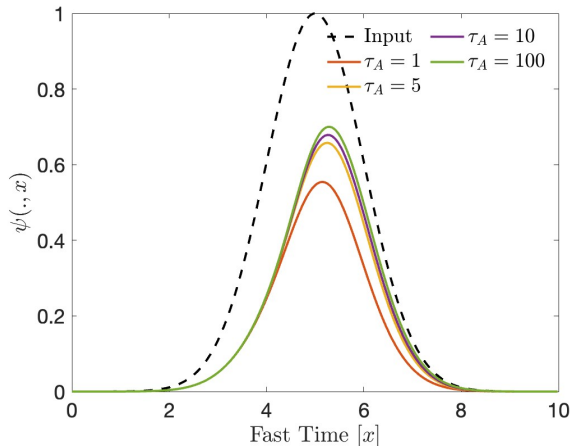
$$\frac{dn}{dx} = \frac{1-n}{\tau_A} - \frac{|\psi_{\text{in}}|^2}{w_A}n, \quad n(x = -\infty) = 1, \quad (1)$$

where

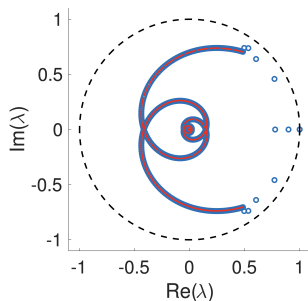
$$\begin{aligned} n(\psi_{\text{in}}) = & \exp\left(\frac{w(x_0) - w(x)}{w_A} + \frac{x_0 - x}{\tau_A}\right) \\ & - \frac{1}{\tau_A} \int_x^{x_0} \exp\left(\frac{w(x') - w(x)}{w_A} + \frac{x' - x}{\tau_A}\right) dx', \end{aligned} \quad (2)$$

is a fraction of the population in the lower level of our SESAM model, and $w(x) = \int_{-\infty}^x |\psi_{\text{in}}(x')|^2 dx'$.

SESAM vs Fast saturable absorber



Stability of the pulse



$\sigma_{\text{ess}}(\mathcal{M}) = \{\lambda_{\pm}(\omega) \in \mathbb{C} \mid \omega \in \mathbb{R}\} \cup \{0\}$, where $\lambda_{\pm}(\omega)$ is given by⁴

$$l_{\text{oc}}(1 - l_0) \exp \left\{ \frac{1}{2} \left(1 - \frac{\omega^2}{\Omega_g^2} \right) \int_0^{L_{\text{FA}}} g(\psi) dt \right\} \exp \left\{ \pm i \left(\frac{\beta_{\text{RT}}}{2} \omega^2 - \theta \right) \right\}.$$

⁴Shinglot V, Zweck J. The essential spectrum of periodically stationary pulses in lumped models of short-pulse fiber lasers. Stud Appl Math. 2023; 150: 218–253.

Thank you!